

ENRIQUE BOTÍA BARBERÁ

Generative AI Engineer · Patent Intelligence & LLM Systems

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PROFILE

Generative AI engineer who designs and ships production LLM systems for patent intelligence. At IP Mind I built PRISM, an end-to-end pipeline that automates Standard Essential Patent (SEP) claim charting against telecom and multimedia standards (3GPP LTE/5G, IEEE 802.11, HEVC), applied at scale to millions of patents across ~20 standards. Background in NLP and deep learning (MSc, UC3M); used to owning a system end to end in a two-engineer team, from architecture and retrieval to evaluation and production.

EXPERIENCE

Generative AI Engineer · IP Mind Ltd

Apr 2024 - Present

Cambridge, UK (remote from Madrid) · Patent-intelligence startup; one of two engineers

- Designed and built PRISM (v1, 2.0, PRISM+), an LLM pipeline that performs SEP claim charting and essentiality assessment automatically, mapping each claim feature to normative passages of the standard with structured verdicts, normativity flags and confidence scores. Applied to millions of patents across ~20 telecom and multimedia standards.
- Reduced claim charting to minutes per claim, a task that traditionally takes patent specialists hours of manual work, turning essentiality analysis into a scalable, repeatable process.
- Architected the multi-phase reasoning flow: claim decomposition into semantic feature blocks, multi-query hybrid retrieval (dense + sparse, per-query weighting), per-feature assessment against a rubric, iterative evidence expansion and contrastive verification.
- Built the RAG ingestion layer for standards specifications: document parsing and chunking (Reducto), table-to-prose representation, embedding-model benchmarking (Voyage, Qwen3, Gemini) and a Pinecone hybrid index.
- Set up LLM-as-a-judge evaluation and JSON-schema structured outputs; eliminated hallucinated cross-references with verbatim anchoring, whitelisted references and deterministic validators, making outputs auditable for patent attorneys.
- Deployed and operate the platform on Azure with n8n orchestration and FastAPI services; own prompt design, agent schemas, debugging on real data and production reliability.

Data Scientist (previously AI Intern) · Icaria Technology

Mar 2023 - Apr 2024

Madrid, Spain

- Built NLP / NER models with spaCy and Hugging Face to detect sensitive data in semi-structured and unstructured text, with results shared through Streamlit apps.
- Developed forecasting models and raised the team's engineering baseline (Python best practices, Git workflows, reproducible ML infrastructure).

Research Assistant · Universidad Carlos III de Madrid

Jan 2022 – Jan 2023

Leganés, Spain

- Developed PyTorch deep-learning models on speech signals for mental-health research (depression detection); literature review and experiment design.

SKILLS

LLM systems: RAG and hybrid retrieval, prompt engineering, agent workflows, structured outputs / JSON schemas, LLM-as-a-judge and evaluation pipelines, hallucination control

Tools: Python, PyTorch, Hugging Face, spaCy, FastAPI, n8n, Pinecone, Azure (Azure OpenAI), OpenAI / Gemini APIs, Streamlit, Git

Domain: Standard Essential Patents, claim charting and essentiality analysis, 3GPP / IEEE / ITU-T standards, patent documents and legal-grade traceability

EDUCATION

MSc in Machine Learning for Health · Universidad Carlos III de Madrid

2022 – 2023

Thesis: detection of depression in speech using pre-trained embedding models

BSc in Data Science & Engineering · Universidad Carlos III de Madrid

2018 – 2022

Thesis: mental-disorder severity estimation from speech analysis and ML

ACHIEVEMENTS & LANGUAGES

Achievements: Finalist, Cajamar UniversityHack 2022 Datathon · Udacity NLP Nanodegree (2021) and Programming for Data Science Nanodegree (2020)

Languages: Spanish (native) · English (C1, working language at IP Mind)